

Total No. of Questions : 8]

SEAT No. :

PE-2584

[Total No. of Pages : 2

[6583]-115

T.E. (Electronics & Computer Engineering)
MICROCONTROLLER & APPLICATIONS
(2019 Pattern) (Semester - V) (310344)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) Draw interfacing diagram and write a C program for the 8051 microcontroller to transfer the letter "A" serially at 4800 baud continuously. Use 8 bit data and 1 stop bit. **[6]**
- b) Draw interfacing diagram of DAC to 8051 microcontroller and write an Embedded C program to generate sine wave form. **[6]**
- c) Explain the working principle of motion detector (PIR) sensor. Draw interfacing diagram of motion detector sensor and buzzer with 8051 microcontroller and write a Embedded C program to turn on buzzer when motion is detected. **[8]**

OR

- Q2)** a) Interface buzzer with 8051 microcontroller. Write an Embedded C program for the same. **[6]**
- b) Draw interfacing diagram of opto - isolator with 8051 microcontroller. Also write an Embedded C program for same. **[6]**
- c) Draw interfacing diagram of stepper motor with 8051 microcontroller. Write an Embedded C program to rotate stepper motor continuously in clockwise direction. **[8]**

- Q3)** a) List features and applications of variants of MSP430 family MSP430x2x, MSP430x4x, MSP430x5x. **[8]**
- b) Explain in detail: **[8]**
- i) Low power modes in MSP430 microcontroller.
 - ii) Compare FRAM and Flash memory.

OR

P.T.O.

- Q4)** a) State the function of following registers of MSP430 microcontroller.[8]
i) Status Register
ii) Program Counter
iii) Stack pointer
b) Explain in detail any four addressing modes of MSP430 microcontroller.[8]

- Q5)** a) Write an Embedded C program for blinking of LED with and without switch using MSP430. Also draw its interfacing diagram. [8]
b) Interface relay with MSP430 microcontroller. Write an Embedded C program for the same and draw its flowchart. [8]

OR

- Q6)** a) Draw interfacing diagram of IR sensor with MSP430 microcontroller. Also write Embedded C program for the same along with its flowchart.[8]
b) Draw and explain interrupt structure of MSP430. [8]

- Q7)** a) Explain with neat block diagram DAS using 8051 microcontroller. [9]
b) Design environment Monitoring System using MSP430 microcontroller. Also draw the flowchart for same. [9]

OR

- Q8)** a) Design Home Automation System using MSP430 microcontroller. Also draw the flowchart for same. [9]
b) Design frequency counter using 8051 microcontroller which will display the result of various frequency ranges on LCD. Also write the algorithm for same. [9]

